# SuperMimic on Kangaroo, M5: the 24 h `full` and `dev` runs

The paper's algorithm with the M4 stability fixes, on all 124,077 BONES-SEED motions and on a 2.5 h subset, one A100 each. Branch `omm/supermimic` of pal_mjlab, 27 October 2026.

## Summary


- **Both runs trained without a collapse of their own.** `full`'s first job went NaN at 17.5 h because the simulator's constraint buffer overflowed (D44); resumed with a larger buffer, it ran 24 h more with no overflow. Its only other incident is a 1,400-iteration dip at 82,700, from which it recovered by itself.
- **On `dev`'s own clips, `dev` is the best policy:** 0 failed clips of 1,294 and tracking return 0.906 at 70,000 iterations, above every earlier result (PPO-Sim 0.770, the stability runs' best 0.894). `full` keeps improving slowly on those clips, to 0.894 with 0 failures at 110,000.
- **On motions `dev` never saw, `full` is the better policy.** On a second 2.5 h sample of the library, `dev`'s final policy fails 23 of 1,242 clips (0.851); every `full` checkpoint fails 1 (0.862–0.878). One `full` policy handles the whole library about as well as `dev` handles its own 2.5 h.
- **The last checkpoint is not the best.** `full`'s last checkpoint (121,000) scores 0.874 with 3 failures, against 0.889 with none at 120,000. Both jobs end with the learning rate still at 54–82% of its start (the paper decays it over 250,000 iterations), so consecutive checkpoints still move around.
- **The dip followed the sampler onto physically impossible references.** Before it, four kneeling clips, whose retargeted knee is 17.7 cm inside the floor in every frame, held 14% of all starts. Such clips can never be tracked, so the hard sampler keeps feeding them. Since then: a dataset QA rule drops clips with any collision shape more than 5 cm below the floor, and the 7-DoF data is being re-retargeted with the GMR fixes of the 5-DoF set (D47).


## 1 The runs

Both jobs ran the same commit on one A100 each: the paper's settings (TF32, replay 1000), compiled imagination (D37), and the stability defaults of M4 (D39: the failure head on the rule's two errors and the survival-weighted critic loss), on the reviewed dataset (D40).

| run | job | data | result |
|---|---|---|---|
| `dev-24h` | 62300428, `e6f3893` | `dev`: 1,294 clips, 2.55 h | finished: 71,910 iterations in 24 h |
| `full-24h` | 62300423, `e6f3893` | `full`: 124,077 clips, 250.9 h | NaN at 44,620 (17.5 h); cancelled |
| `full-24h`, resumed | 62362613, `bc44e76` | `full`, from `model_44000`, `njmax` 1024 | finished: 121,870 iterations, 1.11 s each |

`full` trained about 40 h in all: 15.9 h up to iteration 44,000 in its first job, then 24 h resumed. ONNX policies for the robot (D48) are exported for `dev` 71,000 and `full` 121,000, and both run in the dashboards.

## 2 Benchmark on `dev`

Every checkpoint at a multiple of 2000 (and `full`'s last) plays every clip of the reviewed 1,294-clip `dev` once from its first frame (paper F.1). `full` is drawn as one curve: its first job up to 44,000, then the resumed job.

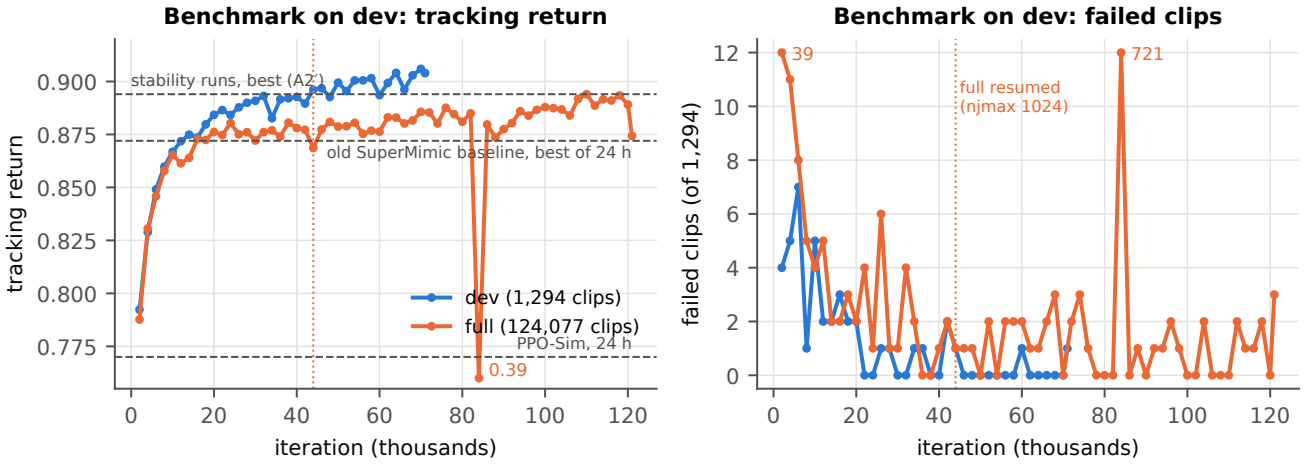



**Figure 1:** Benchmark on `dev`. Dashed lines: earlier results on the 1,296-clip `dev` (it also holds the two clips removed in D40, which every policy fails). Dotted: `full` resumed from 44,000. Points outside an axis are marked at its edge with their value (84,000: inside the dip).

**Table 1:** Selected checkpoints on `dev`. Body error: root-relative mean position error per body group.

| checkpoint | failed clips | tracking return | body error arms / legs / torso |
|---|---|---|---|
| `dev`, 22,000 (7.4 h) | 0 | 0.887 | 5.9 / 4.0 / 1.7 cm |
| `dev`, **70,000** (23.6 h, best) | **0** | **0.906** | 5.4 / 3.6 / 1.4 cm |
| `dev`, 71,000 (last) | 1 | 0.904 | 5.6 / 3.6 / 1.4 cm |
| `full`, 44,000 (resumed from here) | 1 | 0.869 | 6.1 / 4.2 / 1.6 cm |
| `full`, 84,000 (inside the dip) | 721 | 0.389 | 23.7 / 11.5 / 5.1 cm |
| `full`, **110,000** (best) | **0** | **0.894** | 5.5 / 3.8 / 1.3 cm |
| `full`, 120,000 | 0 | 0.889 | 5.5 / 4.2 / 1.4 cm |
| `full`, 121,000 (last) | 3 | 0.874 | 5.6 / 3.5 / 1.4 cm |
| PPO-Sim, 24 h (M4, 1,296 clips) | 15 | 0.770 | 10.7 cm arms |
| old SuperMimic baseline, best of 24 h (M4, 1,296 clips) | 2 | 0.872 | 6.2 cm arms |
| stability runs, best (M4, 1,296 clips) | 2 | 0.894 | |

- `dev` **kept improving for its whole 24 h:** no failed clip from 46,000 on (one checkpoint with 1), from 0.887 at 22,000 to 0.906 at 70,000.
- `full` **improves slowly after the resume:** 0.877 at 46,000, 0.885–0.888 at 70,000–82,000, 0.887–0.894 from 98,000 on, with 0–3 failures. It stays below `dev` on these clips: `dev` trains on each of them about 100 times as often.
- **The 84,000 checkpoint fell inside the dip** (721 failures); 86,000 is back at 0 failures.

## 3 Motions `dev` never saw

A second draw with the `dev` recipe (2.5 h stratified by category, seed 1): 1,242 clips, 14 of them also in `dev`, all of them in `full`'s training data.

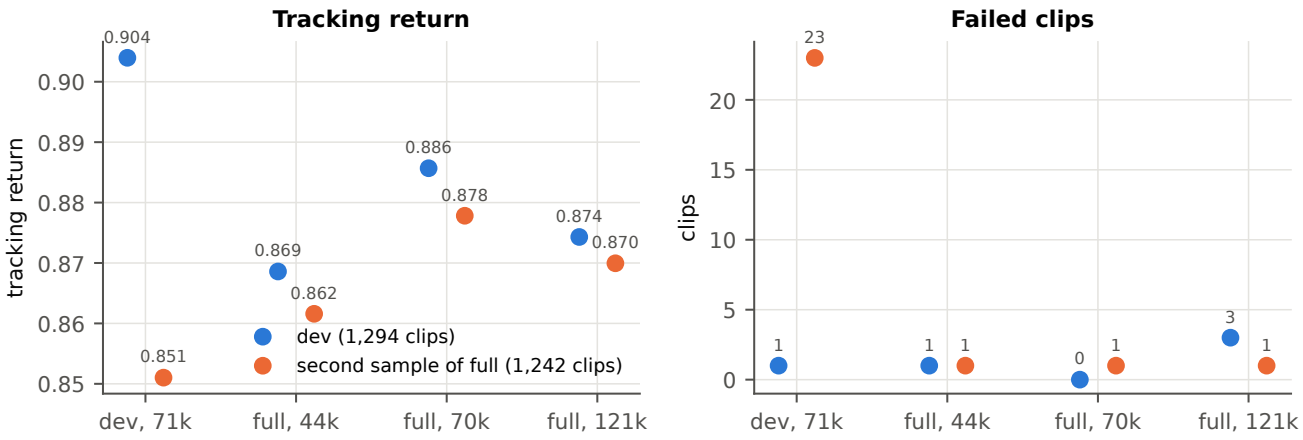



**Figure 2:** The final policies on `dev` (blue) and on the second sample (orange).

**Table 2:** Failed clips and tracking return on both samples; body error on the second sample.

| policy | on `dev` (1,294 clips) | on the second sample (1,242 clips) | body error arms / legs / torso |
|---|---|---|---|
| `dev`, 71,000 | 1, 0.904 | 23, 0.851 | 6.8 / 4.3 / 1.7 cm |
| `full`, 44,000 | 1, 0.869 | 1, 0.862 | 6.3 / 4.3 / 1.6 cm |
| `full`, 70,000 | 0, 0.886 | 1, 0.878 | 5.7 / 3.9 / 1.4 cm |
| `full`, 121,000 | 3, 0.874 | 1, 0.870 | 5.7 / 3.5 / 1.4 cm |

- `dev` **does not transfer to new motions:** 23 failures (1.9%) and 0.851 on motions it never trained on.
- `full` **scores the same on both samples**, within 0.01, and fails 1 clip at every checkpoint: it has trained on all of them.
- The 250.9 h of `full` buy coverage, not a better score on a small set.

# 4 Training signals

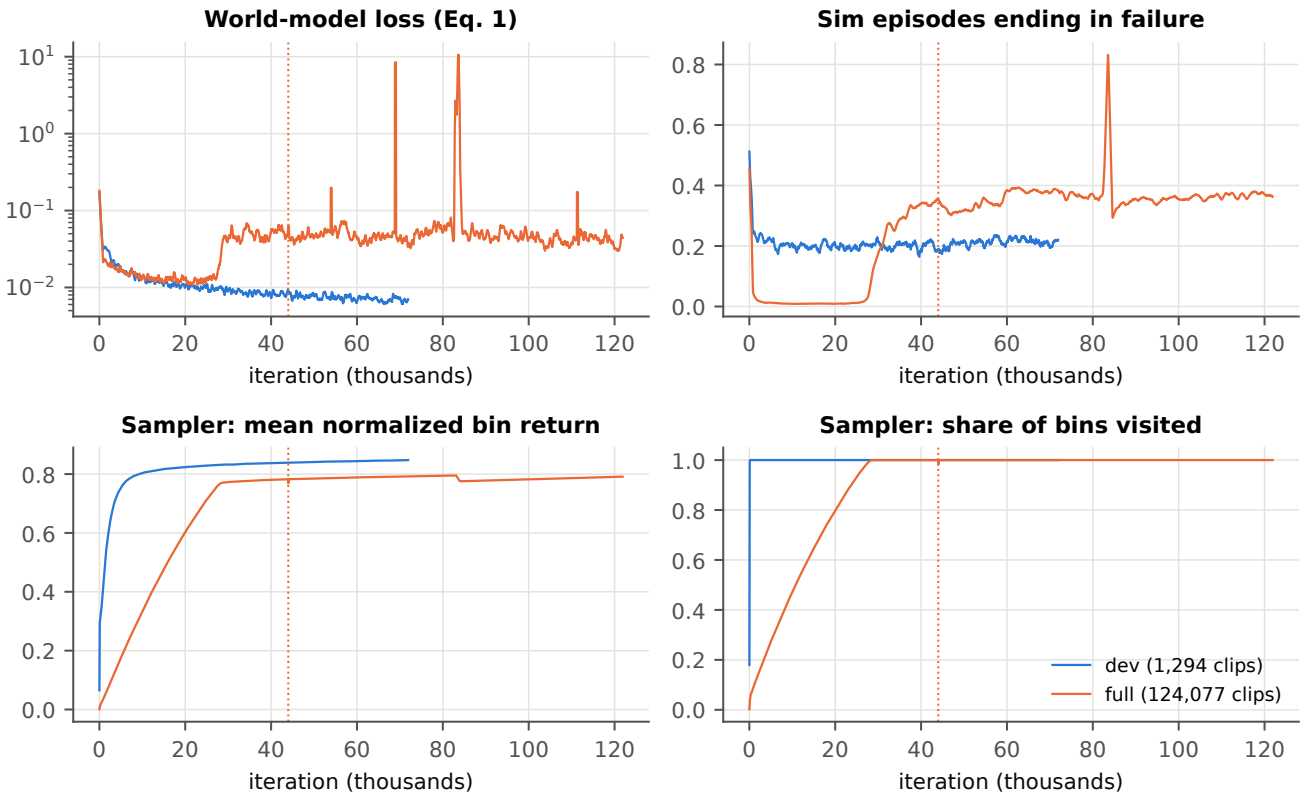



**Figure 3:** TensorBoard signals. World-model loss and the share of failing episodes are moving averages. The sampler's mean return is over all bins, unvisited ones counting as 0.

- `full`**'s sampler had visited all of its 964,607 bins by about 29,000** iterations. From then on, half of its starts go to the hardest 1% of bins (by design: 40% softmax($-25G$) and 10% past weight). So its training failure rose from 1% to 34–39%, and its world-model loss from 0.013 to about 0.045. Neither is a regression: the mean normalized bin return kept rising, 0.766 at 28,000, 0.783 at 44,000, 0.796 at 82,000, and 0.791 at the end (after the dip). Training return is not a progress measure on `full`; the mean bin return and the benchmark are.
- `dev`**:** training failure flat at 21–22% (its sampler always has the hardest clips at hand), world-model loss 0.010 $\rightarrow$ 0.007, mean bin return 0.80 $\rightarrow$ 0.85.
- **Stability:** imagined and simulated 32-step returns correlate at $\geq 0.998$ over the last 5,000 iterations of both runs. `dev`'s critic loss stays below 0.015 after 3,000 iterations. `full` had one critic-loss burst (297, in the dip) and world-model loss spikes above 0.15 at 30,500, 41,500, 46,500, 53,500, 68,500, 81,500–84,000 and 111,000.
- **Memory:** 33.4 GB (`full`), 19.6 GB (`dev`).

# 5 Two incidents on `full`

## The NaN at 17.5 h: the constraint buffer (D44)

At iteration 44,620 the world model, critic and actor went NaN within 10 iterations. All run long, MuJoCo Warp had warned "nefc overflow" about 250 times per 100 iterations (up to 636 constraint rows requested, 300 allowed): dropped rows give wrong contact physics, and one non-finite state in the replay poisons every network. `dev` warned 263 times in 24 h. With `njmax` 1024 the resumed job never overflowed. The training logger skips NaN values, so the collapse showed as missing losses and a jump of training failure to 96%.

### The dip at 82,700: impossible references in the hard bins

- **What happened:** the world-model loss spiked first, then the policy: training failure 79%, critic loss 297, imagined-vs-simulated correlation below 0.9. By 84,100 it had recovered by itself.
- **Where the sampler had put its starts** (its state is in every checkpoint): at 80,000–82,000, four clips held about 14% of all starts, `kneeling_loop_102/103__A064` and their mirrors (3.3–3.8% each). After the dip the hard bins spread out (top clip 0.7%), then `crossed_legs_R_001__A428`–`A432` took the top places (3–4% each).
- **Those clips are physically impossible in the 7-DoF data** (pal_mjlab's collision model): in `kneeling_loop` the right knee is 17.4–17.7 cm in the floor in every frame; in `crossed_legs` the legs pass 9–14 cm through each other and arms and grippers go 16–73 cm into the floor. They had all passed the dataset QA, which checks link origins only (paper E: −12 cm once, −8 cm sustained).
- **Likely mechanism (one occurrence):** a bin that can never be tracked never earns its return, so the hard sampler keeps feeding it. The replay fills with failing, contact-rich data from a handful of motions; the world model's fit degrades there, and the policy follows the wrong model.
- **Consequences** (user, 27 October): the dataset QA now drops clips with any collision shape more than 5 cm below the floor, and the 7-DoF data is re-retargeted with the 5-DoF set's GMR fixes: self-collision avoidance, the clip's own floor, the torso weight, GMR's IK limits, and pal_mjlab's grippers with a physically paired wrist target (D47). In the current 7-DoF data, 30.6% of frames have a self-penetration over 1 cm and 28% a foot over 1 cm in the floor (1,000 random clips).

## 6 The last checkpoint is not the best

- `full` 121,000 is worse than 120,000 on most clips (per clip, median −0.007; 63% of clips worse) and fails 3, where 120,000 fails none. Consecutive checkpoints usually differ by ±0.003–0.005 per clip. On the second sample, 121,000 (0.870) is also below 70,000 (0.878). `dev`'s last checkpoint (71,000: 1 failure, 0.904) is also just below its best (70,000: 0, 0.906).
- **The learning rate is still high when the jobs end.** The paper's cosine schedule decays to 3% over 250,000 iterations; `full` stops at 121,870 with the rate at 54% of its start, `dev` at 71,910 with 82%. So checkpoints keep moving, and the last one is a sample of that noise.
- **To make the last checkpoint the best:** decay the learning rate over the run's own length, or keep an average of the policy weights. Not done yet.

## 7 Caveats

- One seed per run.
- The benchmark runs with `njmax` 1024; `dev` and `full`'s first job trained with 300, so their contacts are simulated more exactly here than in training.
- A checkpoint takes 5.2 min on the local GPU, about 4× the M4 benchmarks; the larger constraint buffer is the likely cause (not measured).

## 8 Decisions and next steps

- **D44:** `njmax` 1024. **D48:** ONNX policies at every checkpoint, and a script for older ones. **D47** (running): the 7-DoF data re-retargeted and re-QA'd with the floor rule; it replaces the current set, which is archived.
- **Next:**
  - a `full` run on the new 7-DoF data, benchmarked on a sample of it;
  - a learning-rate schedule fitted to the run length, or a weight average, so the last checkpoint is the best (needs the user's go);
  - the 5-DoF `full` run (job 62433262) is queued on Leonardo;
  - option: an upper-body goal interface on a frozen checkpoint (paper Appendix B).

# A Reproduction

```
# runs (Leonardo, lazyhpc run folders 20261026-015332_{SMfull,SMdev}_e6f3893, 20261026-192504_SMfull2_bc44e76)
python -m pal_mjlab.scripts.supermimic.train Mjlab-SuperMimic-Flat-Pal-Kangaroo-Grippers-ArmWaist1Hz \
  --library <dataset>/libraries/{full,dev} --agent.experiment-name supermimic_d39 --agent.run-name {full,dev}-24h
# benchmark: dev, and the second sample (bones_to_library --subset dev --seed 1)
python -m pal_mjlab.scripts.supermimic.eval --task <task> --library <library> \
  --checkpoints <run>/model_*.pt --out-dir logs/supermimic_eval/m5_<name>
# ONNX for older checkpoints (D48)
python -m pal_mjlab.scripts.supermimic.export_onnx <task> --library <library> <run>/model_N.pt
# figures and this report
python docs/supermimic/m5/make_figures.py && (cd docs/supermimic/m5 && latexmk -pdf report.tex)
```